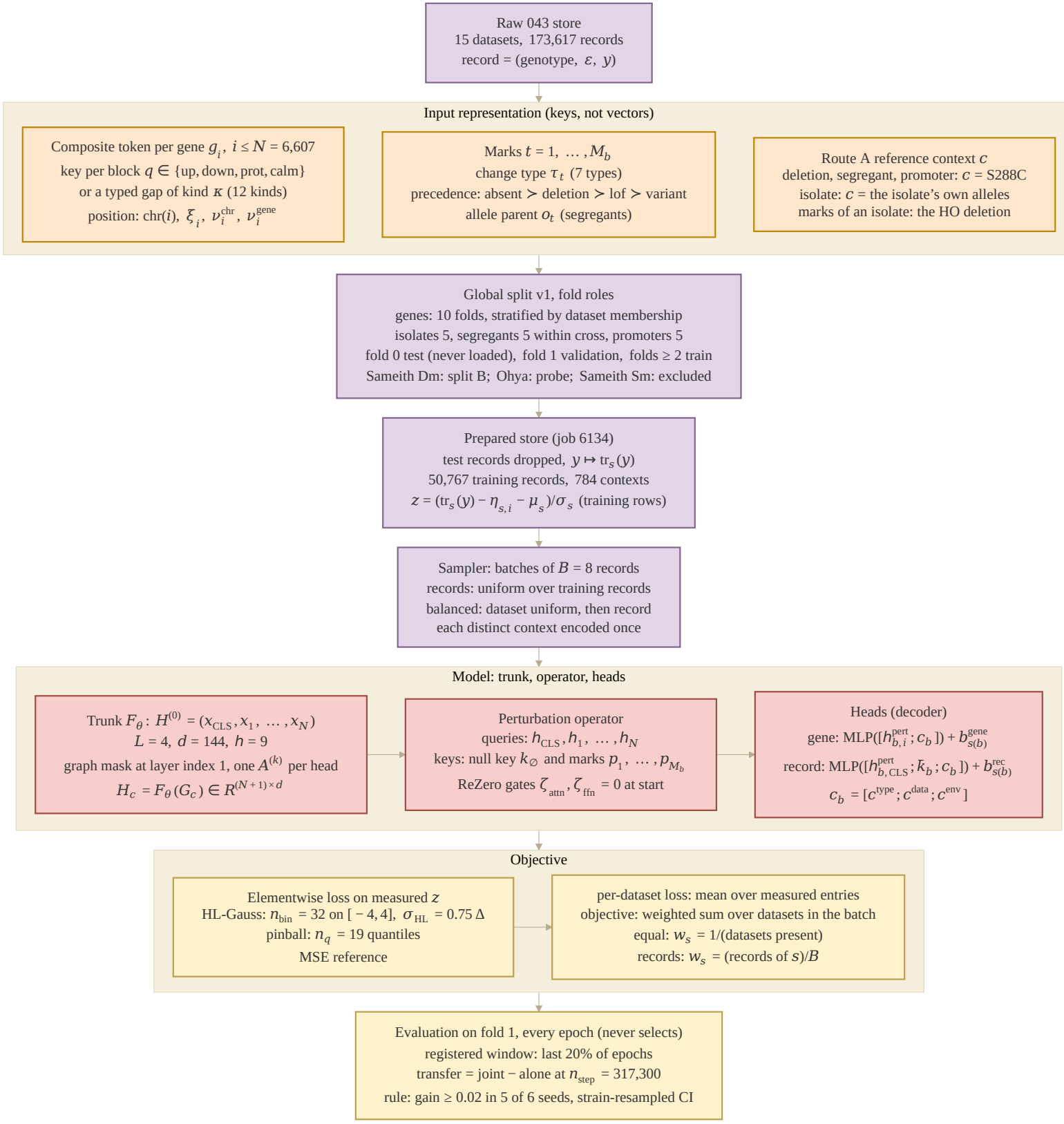

Raw 043 store
15 datasets, 173,617 records
record = (genotype, ε, y)
Input representation (keys, not vectors)
Composite token per gene g_i, $i \le N$ = 6,607
key per block $q \in$ {up, down, prot, calm}
or a typed gap of kind κ (12 kinds)
position: chr(i), ξ_i, ν_i^{chr}, ν_i^{gene}
Marks t = 1, … , M_b
change type τ_t (7 types)
precedence: absent ≻ deletion ≻ lof ≻ variant
allele parent o_t (segregants)
Route A reference context c
deletion, segregant, promoter: c = S288C
isolate: c = the isolate's own alleles
marks of an isolate: the HO deletion
Global split v1, fold roles
genes: 10 folds, stratified by dataset membership
isolates 5, segregants 5 within cross, promoters 5
fold 0 test (never loaded), fold 1 validation, folds ≥ 2 train
Sameith Dm: split B; Ohya: probe; Sameith Sm: excluded
Prepared store (job 6134)
test records dropped, $y \mapsto \mathrm{tr}_s(y)$
50,767 training records, 784 contexts
$z = (\mathrm{tr}_s(y) - \eta_{s,i} - \mu_s)/\sigma_s$ (training rows)
Sampler: batches of B = 8 records
records: uniform over training records
balanced: dataset uniform, then record
each distinct context encoded once
Model: trunk, operator, heads
Trunk F_θ: $H^{(0)} = (x_{\mathrm{CLS}}, x_1, \dots, x_N)$
L = 4, d = 144, h = 9
graph mask at layer index 1, one $A^{(k)}$ per head
$H_c = F_\theta(G_c) \in R^{(N+1)\times d}$
Perturbation operator
queries: $h_{\mathrm{CLS}}, h_1, \dots, h_N$
keys: null key $k_\varnothing$ and marks $p_1, \dots, p_{M_b}$
ReZero gates $\zeta_{\mathrm{attn}}, \zeta_{\mathrm{ffn}}$ = 0 at start
Heads (decoder)
gene: MLP($[h^{\mathrm{pert}}_{b,i}; c_b]$) + $b^{\mathrm{gene}}_{s(b)}$
record: MLP($[h^{\mathrm{pert}}_{b,\mathrm{CLS}}; \bar{k}_b; c_b]$) + $b^{\mathrm{rec}}_{s(b)}$
$c_b = [c^{\mathrm{type}}; c^{\mathrm{data}}; c^{\mathrm{env}}]$
Objective
Elementwise loss on measured z
HL-Gauss: n_{bin} = 32 on [− 4, 4], σ_{HL} = 0.75 Δ
pinball: n_q = 19 quantiles
MSE reference
per-dataset loss: mean over measured entries
objective: weighted sum over datasets in the batch
equal: w_s = 1/(datasets present)
records: w_s = (records of s)/B
Evaluation on fold 1, every epoch (never selects)
registered window: last 20% of epochs
transfer = joint − alone at n_{step} = 317,300
rule: gain ≥ 0.02 in 5 of 6 seeds, strain-resampled CI